

Retail Sales Analytics & Forecasting

Historical insights, store comparisons and forecasting across a 45-store network

SALES RECORDS

421,570

STORES

45

SALES TOTAL

\$6.74B

HOLIDAY UPLIFT

+7.84%

BUSINESS QUESTION

How do sales vary across the calendar, holidays and store characteristics—and can historical patterns improve short-term store-department forecasts?

ANALYTICAL WORKFLOW

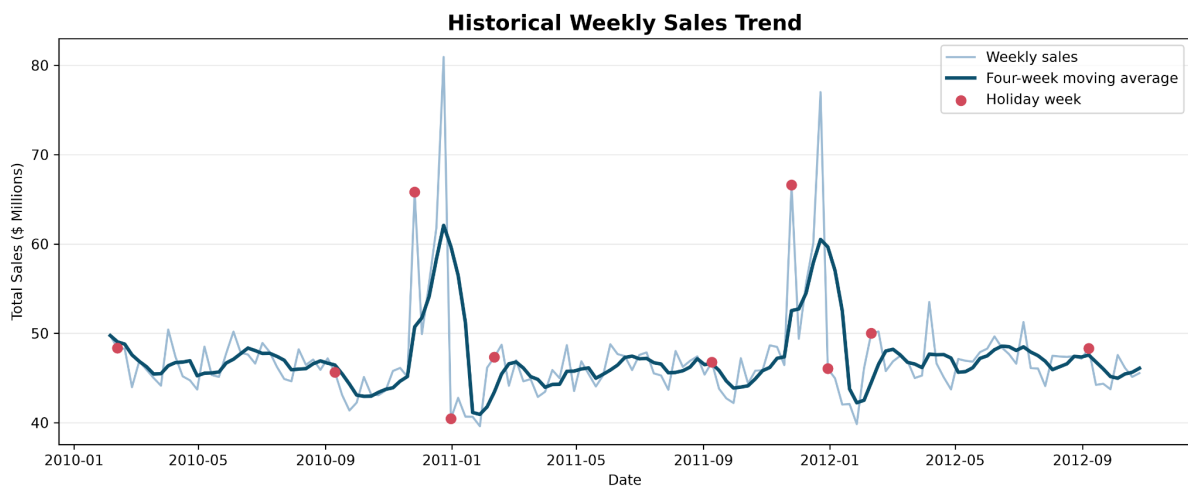
01 Inspect data

02 Clean & merge

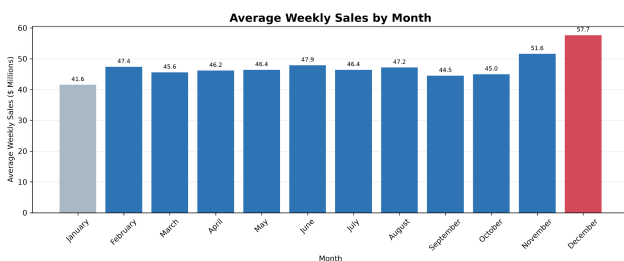
03 Visualise

04 Evaluate models

COMPANY-WIDE WEEKLY SALES | FEBRUARY 2010 - OCTOBER 2012

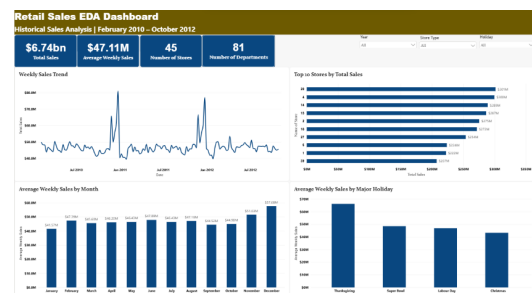


MONTHLY SEASONALITY



December: \$57.7m/wk | January: \$41.6m/wk

POWER BI REPORT SNAPSHOT



Exploratory visuals and interactive filtering

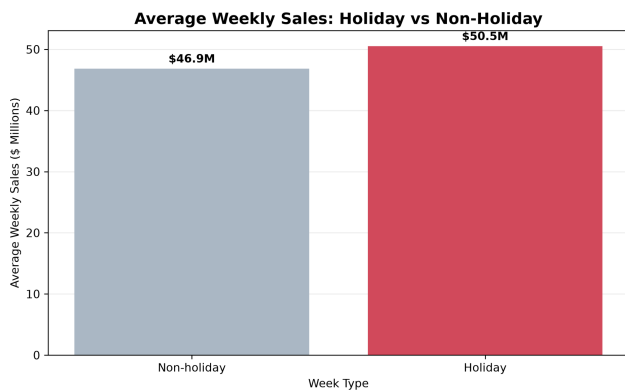
Holiday & Store Performance

The figures describe historic patterns, not guaranteed causes or future outcomes.

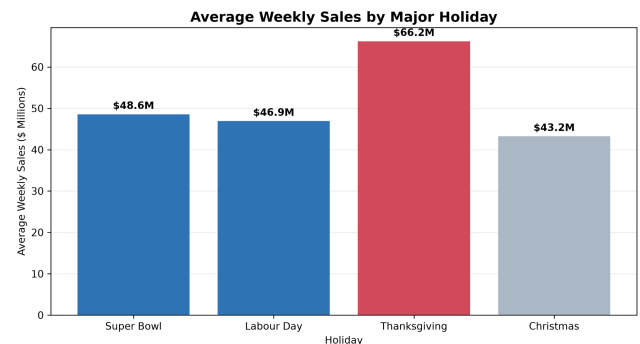
HOLIDAY PATTERNS

Holiday weeks averaged \$50.53m versus \$46.86m in other weeks. Thanksgiving had the highest mean among four named holidays, but each holiday had only two or three observed weeks.

HOLIDAY VS NON-HOLIDAY



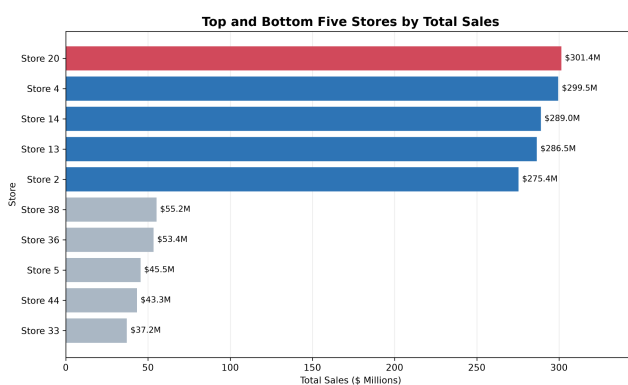
MAJOR HOLIDAY COMPARISON



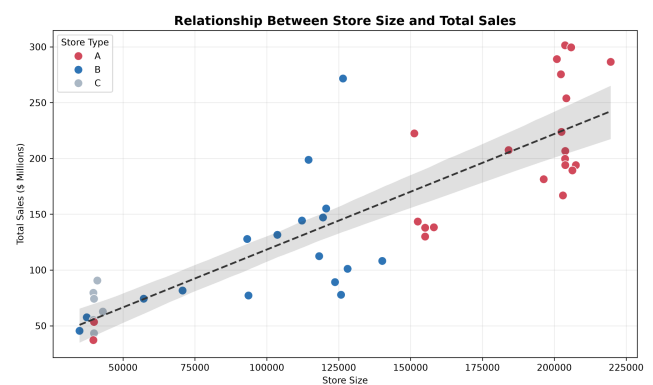
STORE COMPARISONS

Store 20 led at \$301.4m in cumulative sales; Store 33 recorded \$37.2m. Store type and physical size were associated with differences in total sales.

TOP / BOTTOM FIVE STORES



SIZE VS CUMULATIVE SALES



INTERPRETATION & CAVEAT

Store size and cumulative sales were positively correlated ($r \approx 0.846$). That relationship does not prove store size drives revenue. The 10 holiday weeks also limit confidence in individual holiday averages.

Turning Trends Into Forecasts

A chronological holdout evaluates forecasting on later observations, rather than mixing past and future data.

FORECASTING METHOD

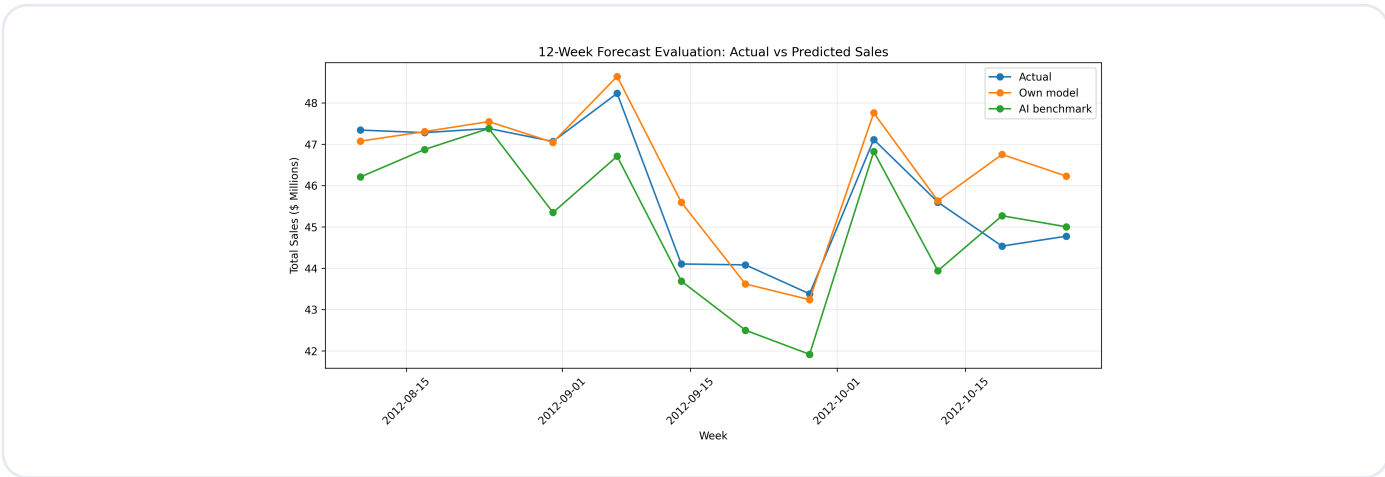
Histogram Gradient Boosting used store and department characteristics, calendar and markdown information, exact 1/4/13/52-week sales lags and shifted rolling means. A 52-week seasonal-naïve rule provided a transparent benchmark.

TIME-ORDERED HOLDOUT
10 Aug–26 Oct 2012 · 12 weeks · 32,938 eligible store–department observations · Same test rows for both models

MODEL EVALUATION

METRIC	GRADIENT BOOSTING	52-WEEK BASELINE
MAE	\$1,294.90	\$1,778.68
RMSE	\$2,696.99	\$3,752.05
Holiday-weighted MAE	\$1,344.84	\$1,811.82
WAPE	7.74%	10.63%

OBSERVED VS PREDICTED AGGREGATE WEEKLY SALES



BUSINESS USE & LIMITATIONS
Use historic trends to prioritise inventory and staffing reviews around peaks, then validate decisions with fresh data. The observed holdout favoured Gradient Boosting, but holiday samples were small, markdown effects are not causal, and forecasts after October 2012 have no observed values for validation.